

Boron-functionalized carbon felt electrode for enhancing the electrochemical performance of vanadium redox flow batteries

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ABSTRACT

Herein, boron-doped carbon felt (B-CF) was proposed as a promising electrode for ensuring the outstanding electrochemical performance of vanadium redox flow batteries (VRFBs). Boron-functional groups can be successfully introduced onto carbon felt (CF) via NH_4BF_4 treatment. The hydrophobic nature of CF is changed to hydrophilicity via the incorporation of boron-functional groups, resulting in excellent affinity with the electrolytes of VRFBs. A VRFB cell containing B-CF exhibits higher energy efficiency (80.56%) than that of CF (63.40%) at 100 mA cm^{-2} current density. This improvement is attributed to the enhanced electrocatalytic properties of B-CF toward $\text{VO}_2^+/\text{VO}^{2+}$ and $\text{V}^{2+}/\text{V}^{3+}$ redox reactions.

1. Introduction

The electrode is one of the most important components of the vanadium redox flow battery (VRFB) because it offers active sites that enhance the electrochemical activity for $\text{VO}_2^+/\text{VO}^{2+}$ and $\text{V}^{2+}/\text{V}^{3+}$ redox reactions. In general, carbon felts (CFs) have been used as electrodes for VRFBs [1]. The electrochemical performance of CFs primarily relies on their physicochemical properties, such as their surface functionality, wettability, and specific surface area. Therefore, numerous studies have enhanced the electrochemical performance of the CF through the modification of their physicochemical properties [2–5].

The incorporation of heteroatoms, such as nitrogen (N), phosphorus (P), fluorine (F), and nitrogen–phosphorus co-doping in backbone carbonaceous materials, is a powerful strategy for altering the physicochemical properties of CFs [5–7]. Boron is an interesting element for the modification of the surface properties of carbonaceous materials, because boron has a lower electronegativity than carbon and acts as an electron donor; further, the carbonaceous π -electrons in a conjugated system transfer to the free P_z orbital of boron via a reduction reaction [8]. Although boron is an attractive element for the effective alteration of the electrochemical properties of carbon-based materials, boron-doped carbon materials as electrocatalysts for $\text{VO}_2^+/\text{VO}^{2+}$ and $\text{V}^{2+}/\text{V}^{3+}$ redox reactions have rarely been studied in VRFBs [9,10].

Herein, we propose a boron-doped CF (B-CF) prepared via facile chemical treatment using NH_4BF_4 as an electrode for all VRFBs. As

expected, the boron-doped CF enables a change of the hydrophobic nature of the CF surface to superhydrophilicity owing to the introduction of boron-functional groups. Consequently, the highly competitive electrochemical performances of the proposed electrode were demonstrated when applied to VRFBs.

2. Results and discussion

Fig. 1a displays a schematic of the incorporation of the boron-functional groups into the CF electrode via the thermal decomposition of NH_4BF_4 , performed to enhance the electrode electrochemical performance toward the $\text{VO}_2^+/\text{VO}^{2+}$ and $\text{V}^{3+}/\text{V}^{2+}$ redox reactions. According to the energy dispersive X-ray spectroscopy (EDS) and Fourier transform infrared (FT-IR) spectroscopy results (Fig. S1), B atoms were homogeneously detected, indicating that B atoms were introduced onto the CF.

To verify the bonding type on the B-CF, X-ray photoelectron spectroscopy (XPS) analysis was carried out. In Fig. 1b, the B 1s peak at approximately 191 eV was only observed in the XPS spectrum of B-CF. The specific bonding configuration of the boron element was analyzed via the XPS deconvolution of the C 1s, O 1s and B 1s spectra (Fig. 1c–e). The detailed XPS analysis is described in the supporting information. According to the XPS analysis, the boron in B-CF bonded with the carbon and oxygen atoms of the boron-functional groups, such as BCO_2 and BC_2O [11–12]. Meanwhile, according to the microstructural analysis

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(Fig. S2), the proposed surface treatment did not cause significant physical deformation of the CF.

To determine the wettability of the electrodes, the contact angle was measured using the water droplet test. In Fig. 2a and -b, the contact angle of the CF was 122.1° , whereas the contact angle of B-CF was not measurable because of quick water absorption. Additionally, B-CF instantly sank to the bottom of the beaker owing to superhydrophilicity (Fig. 2e), implying that the surface nature of CF was changed from hydrophobic to super-hydrophilic after NH_4BF_4 treatment.

To clarify the effect of the boron-functional groups on the $\text{VO}^{2+}/\text{VO}_2^+$ and $\text{V}^{2+}/\text{V}^{3+}$ redox reactions, cyclic voltammetry (CV) tests were performed and the obtained values are summarized in Table S1. As for the $\text{VO}^{2+}/\text{VO}_2^+$ redox reactions (Fig. 2c, Table S1), the cathodic and anodic peak currents of B-CF were larger than those of CF. The peak potential difference of B-CF was significantly lower than that of CF and the ratio of the anodic peak current to the cathodic peak current of B-CF was approximately 1. These results indicate that B-CF possesses outstanding reversibility for the $\text{VO}^{2+}/\text{VO}_2^+$ redox reaction. For CF, an anodic peak related to the oxidation of V^{2+} to V^{3+} appeared at -0.27 V but a distinct cathodic peak associated with the reduction of V^{3+} to V^{2+} was not observed (Fig. 2d). In contrast, the CV profile of B-CF shows distinct oxidation and reduction peaks (Fig. 2d). Moreover, the hydrogen evolution reaction (HER) and reduction of V^{3+} to V^{2+} are clearly distinguishable. This result implies that the boron-functional groups can accelerate the sluggish redox reaction of the $\text{V}^{2+}/\text{V}^{3+}$ by minimizing the overpotential for the $\text{V}^{2+}/\text{V}^{3+}$ redox reaction, thus allowing for symmetric behavior in the negative electrolyte.

Fig. 2f and g show the Randles-Sevcik plots, obtained from the CV results corresponding to various scan rates (Fig. S3), for B-CF and CF; their slope can be used to calculate the diffusion coefficients. The detailed method of the diffusion coefficient calculations is described in the Supporting Information and the obtained values are summarized in Table S2. As for the $\text{VO}^{2+}/\text{VO}_2^+$ redox reaction, the diffusion coefficients of B-CF are higher than those of the CF. The diffusion coefficient of B-CF in the oxidation of the $\text{V}^{2+}/\text{V}^{3+}$ is higher than that of CF. As for the reduction of $\text{V}^{2+}/\text{V}^{3+}$, the diffusion coefficient of B-CF is approximately

$2.50 - 4.05 \times 10^{-3} \text{ cm}^2 \text{ s}^{-1}$, whereas that of CF could not be calculated because it is difficult to distinguish between the HER and the reduction process of the $\text{V}^{3+}/\text{V}^{2+}$. The increasing diffusion coefficients of B-CF indicate that the mass transfer rate of vanadium ions increases at the interface between the electrode and electrolyte. Moreover, the improved charge transfer rate of B-CF toward the $\text{VO}^{2+}/\text{VO}_2^+$ and $\text{V}^{2+}/\text{V}^{3+}$ redox couples was supported by the electrochemical impedance spectroscopy (EIS) results (Fig. S4). Based on the experimental results and the widely accepted reaction mechanism of VRFBs [1,5], potential reaction mechanisms for the enhanced electrocatalytic activity of B-CF can be proposed (Fig. S5).

To understand the effect of B-CF on the electrochemical performance of VRFBs, a VRFB cell with B-CF (B-CF-cell) and a VRFB cell with CF (CF-cell) were prepared. Each cell was charged and discharged at a current density between 60 and 100 mA cm^{-2} . The B-CF-cell exhibits lower IR drops in the first cycle compared with the CF-cell, indicating that electrochemical polarization can be reduced by adopting B-CF (Fig. 3a). B-CF exhibits higher a voltage efficiency (84.98%) than that of CF (65.92%) at 100 mA cm^{-2} (Fig. 3d). As a result, the B-CF-cell exhibits an energy efficiency of 80.56% at a current density of 100 mA cm^{-2} , which is 15.6% higher than that of the CF-cell (63.40%). In addition, a VRFB cell was charged and discharged with a current density of 60 mA cm^{-2} for 100 cycles. The energy efficiency of the B-CF-cell (84.32%) is higher than that of the CF-cell (73.62%) (Fig. S6). Moreover, the robustness of the boron-functional groups was confirmed by XPS analysis after 100 cycles (Fig. S7). The outstanding electrochemical performance of the B-CF-cell is attributed to the enhanced redox reaction kinetics of the $\text{V}^{2+}/\text{V}^{3+}$ and $\text{VO}^{2+}/\text{VO}_2^+$ redox couples.

3. Conclusions

This study successfully dopes boron onto CF via a facile treatment using NH_4BF_4 . Various analyses indicate that boron doping introduced the boron-functional groups onto the surface of the CF without structural deformation. The resulting boron-doped CF exhibited superhydrophilicity, resulting in outstanding wettability. The B-CF

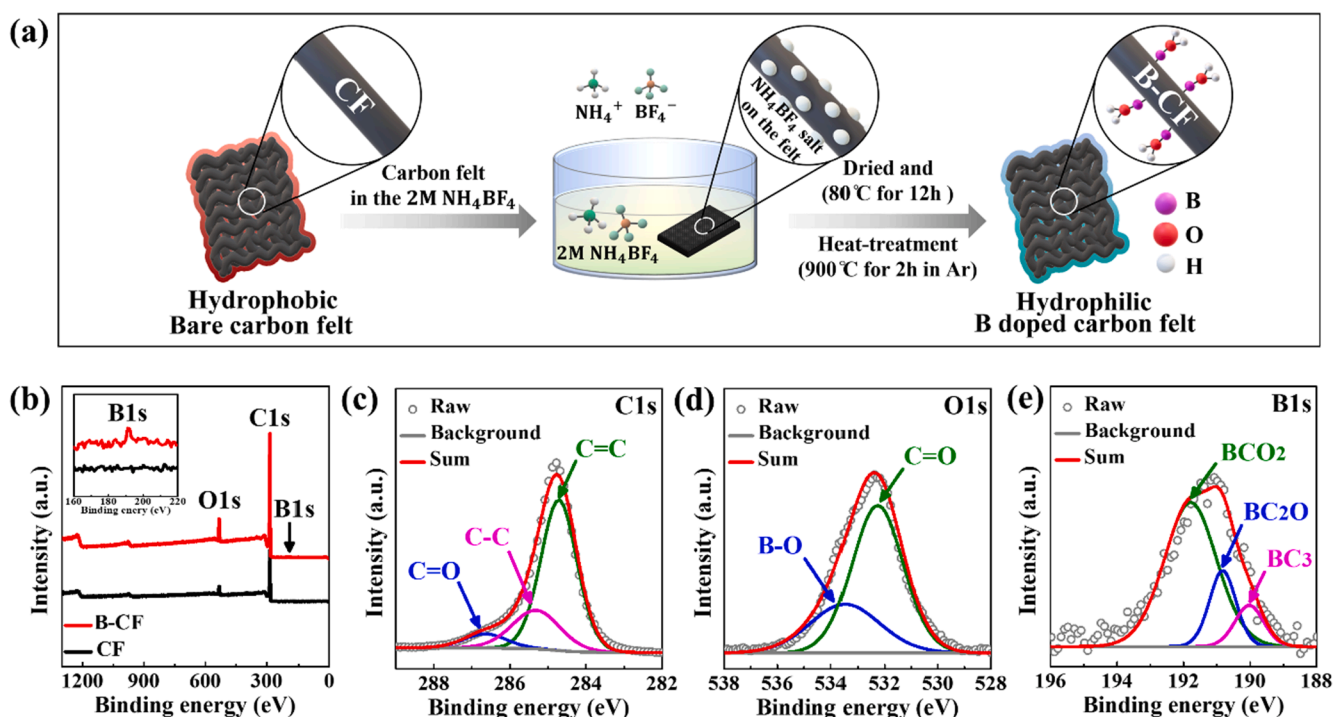


Fig. 1. (a) Schematic of the synthesis of B-CF, indicating the development superhydrophilicity via the introduction of boron-functional groups. (b) XPS survey spectra of B-CF and CF. (c), (d) and (e) correspond to the C1s, O1s, and B1s spectra of B-CF, respectively.

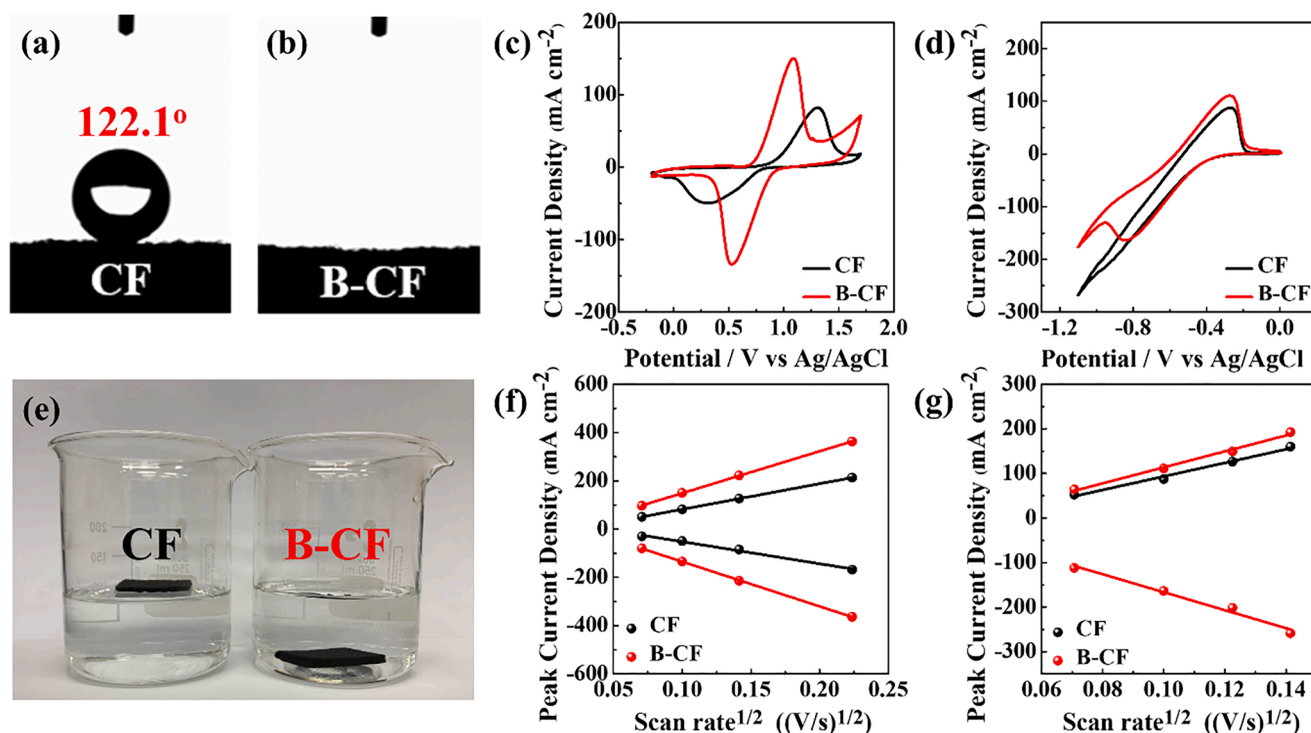


Fig. 2. (a) and (b) show the water droplet tests of CF and B-CF, respectively. (e) shows the soaking test results for CF and B-CF. CV curves of the (c) catholyte and (d) anolyte redox reactions and the peak current densities of the (f) catholyte and (g) anolyte redox reactions.

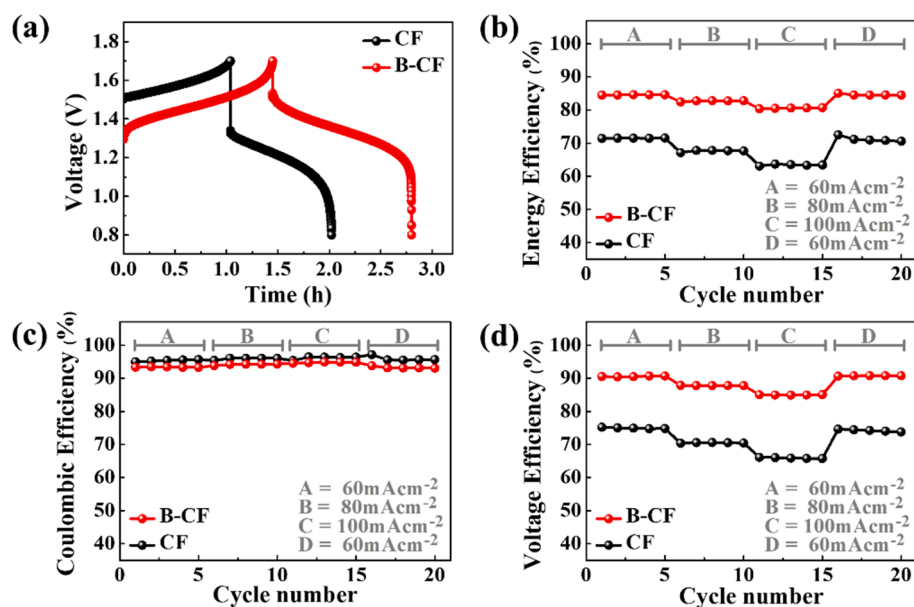


Fig. 3. Electrochemical performance of the VRFB cells at various current densities of 60–100 mA cm⁻²: (a) charge/discharge profile at the first cycle, (b) energy efficiency, (c) Coulombic efficiency, and (d) voltage efficiency.

electrode offers robust active sites for the $\text{VO}_2^+/\text{VO}^{2+}$ and $\text{V}^{2+}/\text{V}^{3+}$ redox reactions, resulting in enhanced mass and charge transfer rates. In particular, the sluggish kinetics of $\text{V}^{2+}/\text{V}^{3+}$ are promoted by the introduction of the boron-functional groups, and consequently, the severely asymmetric behavior in anolyte was significantly alleviated. As a result, the B-CF-cell exhibits superior voltage and energy efficiency accompanied by minimal performance decay over 100 cycles. We believe that the tuning of the electron density in CF through boron-doping serves as a useful guide for the electrode design of VRFBs.

Credit authorship contribution statement

Seong Eun Park: Conception and design of study, Methodology, Acquisition of data, Writing - Original draft preparation. **So Yeon Yang:** Analysis and/or interpretation of data, Investigation, Data curation, Writing - Original draft preparation. **Ki Jae Kim:** Conceptualization, Funding acquisition, Investigation, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apsusc.2021.148941>.

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